

Employee Attrition Prediction – Executive Summary

Project Objective

The objective of this project was to identify employees who are at a higher risk of leaving the organization by analyzing historical HR data. Predicting employee attrition allows HR teams to take preventive actions, reduce hiring costs, improve employee satisfaction, and retain valuable talent.

Key Findings

The analysis showed that employee attrition is influenced by several factors rather than a single reason. Employees with lower monthly income, younger age, and fewer total working years were more likely to leave the company. Employees in the Sales department, especially Sales Representatives, also experienced higher attrition compared to other departments.

A predictive model was developed and multiple machine learning techniques were compared. The selected model successfully identified employees who may be at risk of leaving. The most influential factors included Monthly Income, Age, Total Working Years, Years at Company, Distance from Home, and Overtime.

Business Recommendations

HR should focus on retaining younger and early-career employees by providing mentoring, career development opportunities, competitive compensation, and regular performance discussions. Employees in high-attrition departments should receive additional support through stay interviews and targeted engagement initiatives. While this predictive model provides valuable insights, it should be used alongside managerial judgment because personal reasons for resignation, such as family responsibilities or external job opportunities, are not captured in the available data.